# Modules – Communication

**Menu: Modules > Communication**

Configures how the application communicates with rate control modules. There are three tabs: **Ethernet**, **WiFi** and **CanBus**.

Modules talk to the application either over the network (Ethernet, or WiFi for ESP32 modules) or over CAN Bus. Tick **Enabled** on the Ethernet tab or on the CanBus tab and press Save; ticking one unticks the other. Ethernet is the default. With CAN Bus, Ethernet stays active for AgOpenGPS.

## Ethernet

Chooses the wired network adapter the modules are plugged into, and moves the modules to that adapter's subnet.

| Control | Description |
|---|---|
| **Enabled** | Use the network for module communication (unticks CAN Bus). Applied on Save. |
| **Adapter** | This PC's wired adapters that have an address, shown as name and subnet, e.g. *Ethernet 2 (192.168.1.x)*. WiFi and the tablet hotspot are not listed. If the saved subnet is on no adapter of this PC, such as after a new install or an adapter change, the page picks one by itself: the adapter modules are heard on, otherwise the first. |
| **Status line** | *Modules on this adapter: N*, *No modules heard on this adapter. Press Send to put them on it.*, or *No wired adapter on …* when the cable is unplugged. |
| **Refresh** | Reads the adapter list again, after plugging in a cable or an adapter. |
| **Send** | Tells the modules to move to the selected adapter's subnet. They restart and are counted on the status line once heard. After choosing a different adapter, Send stays greyed out until Save is pressed. |

The steps are shown on the page: select the adapter, Save, then Send.

> **Note:** An ESP32 connected over WiFi ignores Send, so a subnet meant for the cable doesn't take it off its WiFi network.

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## WiFi

Sets up an ESP32 module to join a WiFi network from here, instead of from the module's own web page. Setup also gives the module its ID. The module can join either the tablet's own hotspot or the WiFi network this PC is connected to.

| Control | Description |
|---|---|
| **Tablet Hotspot** | The module joins this tablet's Windows Mobile Hotspot. RateController keeps the hotspot on while this is chosen. It is applied on Save, and the hotspot is started every time RateController starts. The hotspot is set to 2.4 GHz, because the module can't use 5 GHz. |
| **Current WiFi Network** | The module joins the network this PC is connected to in Windows, such as a shop router or a phone hotspot. Connect the PC to it first. |
| **Network** | The network the module will join. For the hotspot: its name and state, which is On, Off, Checking..., *firewall not allowed*, *Off, restarts failed*, or Windows' reason when it can't start. For the PC's network: its name, or *Not connected*. |
| **Hint link** | For the hotspot: *If necessary, turn off Power saving in Hotspot settings*. Windows turns the hotspot off after 5 minutes with no device connected. For the PC's network: a link to **WiFi settings**. |
| **Band line** | The hotspot's band. It also warns when this PC's WiFi is on 5 GHz: the module needs the network's 2.4 GHz band. |
| **Discovered modules** | Module hotspots in range, with signal strength. A module's hotspot is named **RateModule*_ID_6 characters***, e.g. *RateModule_25_1A2B3C*; a new board shows ID 25. Select the module to set up. A lone module is selected by itself. Selecting a module that already has an ID picks that ID, and selecting a new board picks the lowest free ID. The refresh button scans again. |
| **Network password** | The password of the PC's network. RateController remembers the last one used for each network name and fills it in, and the log says so. If the network's password has changed since then, type the new one. Press the eye button to see what's in the box. It isn't needed for the tablet hotspot, whose password the application knows. |
| **Module Password** | The module's own hotspot password. Leave it blank when the module hotspot is open (the default). It is remembered. |
| **Selected Module ID** | The ID the module gets, 0–7. Connected IDs are marked *(Connected)*. The default is the lowest free ID. Picking an ID selects that module's hotspot in the list, if it's there. |
| **Address link** | The web page address of the selected module, when it's a connected ESP32 on WiFi. Click it to open the page. Other boards show *No web page*, and an ESP32 on Ethernet shows *Ethernet, no web page*. |
| **Set Up** | Starts the setup (see below). |
| **Cancel** | Stops a setup. The PC is put back on its own network first. |
| **Log** | Each step of the setup is written in the box. The log button opens the whole log in a larger window. |

### Setting Up a Module

1. Choose **Tablet Hotspot** or **Current WiFi Network**, and press Save if it changed.
2. Power the module and wait for its hotspot to appear in the list. Select it.
3. Check the **Selected Module ID**, and type the network password if needed.
4. Press **Set Up**. The PC leaves its network for a moment, joins the module's hotspot and sends it the network name, password and ID. The module restarts and joins the network. The PC goes back to its network and waits up to a minute for data from the module, then asks the module what went wrong if nothing arrives. The log ends with *Done: module N is connected.* when it worked.

The page asks first if the chosen ID is already connected (only choose it when setting up that same module again), or if the PC's network password is blank (only for an open network). After a successful setup the list is scanned again, and the next free ID is chosen for the next module.

> **Note:** If setup ends with "*Not joined yet*" after the network's password was changed, the old password was probably sent. Some networks, including iPhone hotspots, refuse a wrong password in a way the module

> can't tell apart from a weak signal, so it isn't reported as a refused password. Type the password again and repeat the setup.

### Firewall

Windows Firewall can block module data on a phone hotspot or router network. On the first visit to this tab in a session, and after a setup where no data arrived, the page asks to allow it. Windows then asks for administrator permission. The tablet hotspot needs its own permission the first time it is saved. If that's declined, the Network line shows *firewall not allowed*.

> **Note:** The ESP32's radio is 2.4 GHz only. A network that is 5 GHz only can't be used. A dual-band router works when its 2.4 GHz band has the same name.

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## Module WiFi (ESP32)

ESP32 modules run their own hotspot and settings page. The **WiFi** tab above does the usual setup for you. The page is for checking the module or changing it by hand. Connect a phone, tablet or laptop to the module's hotspot, named **RateModule_*ID*_** followed by six characters unique to that board. Then browse to **192.168.*(200 + module ID)*.1**: module 0 is at 192.168.200.1, module 1 at 192.168.201.1, and a new board (ID 25) at 192.168.225.1. Press **Network**. The hotspot is open unless a hotspot password has been set.

The Network page is where the module is pointed at your own WiFi network, so it can reach the application over an existing network instead of you connecting to the module directly.

| Control | Description |
|---|---|
| **Network** | Name of the WiFi network the module joins |
| **Password** | Password for that network, up to 63 characters (fewer if it uses accented or special characters). The network name can be up to 32 characters. |
| **Use this Network** | Ticked, the module joins the network above. Unticked, the module runs its hotspot only. |
| **Scan for Networks** | Lists the networks in range with signal strength, channel, and a padlock on encrypted ones. Tap a network to fill in its name. |
| **Retry Connection Now** | Forces an immediate connection attempt. Shown only when the module is set to join a network and is not connected. |
| **Hotspot Password** | 8 to 10 characters. Leave empty for an open hotspot. |
| **Save/Restart** | Saves the settings and restarts the module |

> **Note:** Tap a network in the scan results *before* typing the password. Tapping reloads the page, which clears anything already typed.

### Status

| Status | Meaning |
|---|---|
| **Wifi Connected to *name* (address) on channel *n*** | Joined. The address shown is the module's address on your network. |
| **Wifi Not Connected — retrying in the background** | Set to join a network but not connected. The module keeps trying. |
| **Password refused by *name*** | The network rejected the password. Correct it and press Save/Restart. |
| **Hotspot only** | Use this Network is unticked. |

### Joining and Retrying

A module set to join a network keeps trying for as long as the setting is ticked. It does not switch itself back to hotspot-only, and it does not restart itself, so a network that goes down — the shed WiFi switched off, or the machine driving out of range — no longer interrupts work in progress. Earlier firmware gave up after several failures, turned **Use this Network** off and rebooted the module.

Attempts are deliberately spaced out. The module has a single radio shared between its hotspot and its connection to your network, so every attempt takes the hotspot briefly off the air. While a device is connected to the hotspot — a tablet running the application, for instance — attempts are about two minutes apart, or about five minutes if the module has never successfully joined that network. With nothing on the hotspot the module tries more often. Press **Retry Connection Now** rather than waiting.

Loading the module's settings pages pauses the retries for a minute after the last page, so the pages stay responsive while you are configuring. **Retry Connection Now** works regardless.

Scanning for networks takes the radio off the hotspot for a second or two, so a brief pause after pressing **Scan for Networks** is expected.

> **Note:** The hotspot uses the same channel as the network the module joins. Devices on the hotspot drop once the first time a module joins a network, and again if that network later changes channel — they reconnect on their own. From then on the module starts its hotspot on the remembered channel.

Some routers refuse a wrong password in a way the module cannot tell apart from a weak signal. When that happens the status stays on **Wifi Not Connected** instead of naming the password, so check the password whenever a module will not join a network you know is in range.

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## CAN Bus Settings

These settings apply when **Enabled** is ticked on the CanBus tab and saved.

### CAN Adapter

| Driver | Description |
|---|---|
| **SLCAN** | Serial port-based CAN adapter. Requires COM port selection below. |
| **InnoMaker USB2CAN** | Native USB CAN adapter (InnoMaker hardware). No COM port required. |
| **PCAN** | Native USB CAN adapter (Peak PCAN hardware). No COM port required. |

### COM Port (SLCAN only)

1. Use the dropdown to select the COM port the SLCAN adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

### Diagnostics

| Option | Description |
|---|---|
| **Diagnostics** | Enables CAN traffic debug logging in the activity log |

### Status Indicators

| Indicator | Meaning |
|---|---|
| **Module** | Green — module data received within the last 2 seconds |
| **Driver** | Green — CAN adapter is open and receiving frames within the last 4 seconds |

> **Note:** Changing the CAN adapter type or COM port while CAN Bus is active will restart the CAN bridge with the new settings.